

Pierce™ Proteomics Magnetic Clean-up Beads

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WARNING! Read the Safety Data Sheets (SDSs) and follow the handling instructions. Wear appropriate protective eyewear, clothing, and gloves. Safety Data Sheets (SDSs) are available from thermofisher.com/support.

Product description

The Thermo Scientific™ Pierce™ Proteomics Magnetic Clean-up Beads enable fast, efficient, and reproducible removal of a wide variety of detergents, salts, and other mass spectrometry (MS)-incompatible contaminants from protein and peptide samples prior to downstream MS analysis. After binding to the Proteomics Magnetic Clean-up Beads, proteins are enzymatically digested on-bead using MS-grade proteases before reversed-phase peptide fractionation or clean-up. The Proteomics Magnetic Clean-up Beads are compatible with common solvents (e.g. acetonitrile, ethanol, isopropyl alcohol, and acetone) and a wide range of pH (e.g. pH 2-13). The beads can be successfully used to clean up proteins derived from different sample types (e.g., purified proteins, cell lysates, tissue extracts, and biological fluids) and protein input amounts (e.g., 1 µg to 1 mg) using manual magnetic processing or automated workflows.

Contents

Pierce™ Proteomics Magnetic Clean-up Beads (Cat. No. A57868)		
Reagent	Amount	Storage
Magnetic bead slurry (25%)	5 mL	Store at 4°C. Do not freeze.

Additional information

- For maximum protein binding, use a minimum ratio of 1:25, bead slurry (µL):protein (µg) (e.g., 4 µL bead slurry for 100 µg protein sample).
- For processing smaller sample amounts (e.g. 1–25 µg), dilute Proteomics Magnetic Clean-up Beads in (1:10, v:v) in Protein Binding Solution.
- Dilute protein samples (<1 µg/µL) should be concentrated before protein binding with a low MWCO filter or vacuum dry before protein binding.
- Do not dry or freeze the Proteomics Magnetic Clean-up Beads.

Materials required but not supplied

- Low protein binding tubes (e.g., Thermo Scientific™ Low Protein Binding Collection Tubes, Cat. No. [90410](#))
- Heat block or thermal mixer
- Magnetic stand (e.g., DynaMag™-2 Magnet, Cat. No. [12321D](#))
- Lysis buffer (e.g., EasyPep™ Lysis Buffer, Cat. No. [A45735](#))
- Protein Binding solution: 99.5% IPA, 0.5% TFA
- Wash A solution: 100% acetonitrile
- Wash B solution: 70% ethanol, 30% water
- Digestion buffer (e.g., fresh 50 mM TEAB, 50 mM HEPES, or 100 mM ammonium bicarbonate, pH 7-9)
- Trypsin/Lys-C (e.g., Thermo Scientific™ Trypsin/Lys-C Protease Mix, Cat. No. [A41007](#))

Procedure for manual protein clean-up

The protocol below is specifically optimized for use with the Pierce™ Proteomics Magnetic Clean-up Beads. This sample preparation approach is compatible with a wide range of commonly used chemistries and protocols for lysis, protein extraction, reduction and alkylation. Although the Proteomics Magnetic Clean-up Beads can be used as described in published MS sample preparation protocols such as SP3 or PAC (see General References), the protein clean-up and the peptide recovery may be less efficient than our optimized protocol.

Guidelines for manual purification

- An input protein concentration ≥ 1 $\mu\text{g}/\mu\text{L}$ is required for optimal protein binding before clean-up.
- For low protein concentrations, it is recommended to concentrate the sample before protein binding.
- Adjust volumes accordingly for smaller or larger scale purifications.
- Resuspending proteins or protein lysates in a detergent-based lysis buffer such as the EasyPep™ Lysis Buffer is recommended for protein clean-up.
- Reduction and alkylation of cysteines are also recommended before protein clean-up.

Bind protein to beads, digest protein, and clean up peptides

The procedure below is to process 100 μg of protein sample at 2 $\mu\text{g}/\mu\text{L}$.

1. Vortex the beads to ensure total suspension. Transfer appropriate amount of bead slurry based on a 1:25 ratio of bead slurry (μL):protein (μg) (e.g., 4 μL bead slurry for 100 μg protein sample) into a microcentrifuge tube or a 2-mL 96 deep well plate.
2. Collect the beads using a magnetic stand. Remove and discard the storage buffer.
3. Rinse the beads by adding 100 μL Protein Binding solution (99.5% IPA, 0.5% TFA) and vortex briefly. Collect the beads using a magnetic stand. Remove and discard the supernatant.
4. Remove the tube from the magnetic stand, then add 200 μL Protein Binding solution to reconstitute the beads.
5. Combine the reconstituted bead slurry (200 μL) with 50 μL of protein sample for a final 80% organic concentration.
Note: The final protein concentration in the Binding solution is required to be ≥ 0.2 $\mu\text{g}/\mu\text{L}$.
6. Incubate the sample at room temperature for 45 minutes, mixing at 1,200 rpm.
7. Collect the beads using a magnetic stand. Remove and discard the supernatant.
8. Add 150 μL of Wash A solution (100% acetonitrile) and vortex thoroughly. Collect the beads using a magnetic stand. Remove and discard the wash solution.
9. Add 150 μL of Wash B solution (70% ethanol, 30% water) and vortex thoroughly. Collect the beads using a magnetic stand. Remove and discard the wash solution.
10. Add 100 μL of digestion mixture containing 4 μg of trypsin/Lys-C protease mix. Trypsin or Trypsin/Lys-C protease mix quantity reconstituted in the digestion buffer (e.g., 50 mM TEAB, pH 8) should be based on 1:25-1:100 (enzyme:protein (w/w)).
11. Vortex briefly and incubate for a few hours or overnight in a thermal mixer at 37°C, 1200 rpm.
12. After protein digestion, centrifuge briefly to collect any evaporated liquid.
13. Collect the beads using a magnetic stand. Transfer the peptide supernatant to a new tube for C18 peptide clean-up or LC-MS analysis.
Note: We recommend centrifugation of eluted peptide samples at $10,000 \times g$ for 2 minutes before transferring the eluted peptides to avoid bead carryover.

Procedure for automated protein clean-up

Materials required but not supplied

- KingFisher™ Apex with 96 Deep Well Head (Cat. No. [5400930](#)) or KingFisher™ Flex Purification System with 96 Deep-Well Head (Cat. No. [5400630](#))
- KingFisher™ 96 Deep-Well Plate, v-bottom, polypropylene (Cat. No. [95040450](#))
- 96 Deep-Well Tip Combs for KingFisher™ Flex Magnetic Particle Processor (Cat. No. [97002534](#))
- Protein Binding solution, Protein Dilution solution, Wash A solution, and Wash B solution (same as above)
- Heat block or thermal mixer

Guidelines for automated purification

- Load the KingFisher™ Apex instrument with the `Proteomics_Magnetic_Clean-up_100ug.kfx` instrument script (see the *KingFisher™ Apex Purification System User Guide* for details on downloading the script).
- The following protocol is designed for use with the KingFisher™ Apex Magnetic Particle Processor. The protocol can be modified according to your needs directly on the instrument using the touch screen interface or by using the Bindix™ PC Software provided with the instrument.
- If fewer than 96 wells are used in a plate, fill the same wells in each plate. For example, if using wells A1 through A12, use these same wells in all plates.
- To ensure bead homogeneity, mix the vial by repeated inversion, gently vortexing, or rotating the platform before adding the beads to the Bead plate.
- Combine the Tip Comb with a Deep Well 96 Plate.
- See the instrument user manual for detailed instructions.

Prepare instrument and set up plates

1. Dilute 4 μL Magnetic Clean-up Beads in 96 μL Protein Binding solution per sample.
2. Dilute protein sample in Protein Binding solution per sample (e.g., dilute 50 μL , 2 $\mu\text{g}/\mu\text{L}$ of protein sample in 200 μL of Protein Dilution solution).
3. Set up the plate as described in Table 1.

Table 1 Pipetting instructions for the protein purification protocol using Thermo Scientific™ Microtiter DW 96 Plates

Plate #	Plate name	Content	Volume	Time	Mix speed
1	Tip plate	—	—	—	—
2	Bead plate	4 μL Magnetic Clean-up Beads in 96 μL Protein Binding solution	100 μL	—	—
3	Sample plate	50 μL protein (2 $\mu\text{g}/\mu\text{L}$) + 200 μL Protein Binding solution	250 μL	45 minutes	Medium
4	Wash A plate	Wash A solution	150 μL	2 minutes	Medium
5	Wash B plate	Wash B solution	150 μL	2 minutes	Medium
6	Elution/Digestion plate	Digestion mixture	100 μL	4 hours at 37°C	Medium

Perform automated purification

1. On the touchscreen interface, choose the protocol and select the Start button. See the *KingFisher™ Apex Purification System User Guide* for detailed information.
2. Open the door of the instrument.
3. Load plates into the instrument as directed by the touch screen display, placing each plate in the same orientation. Confirm each action on the touch screen.
4. After sample processing, remove the plates as directed by the touchscreen display.

Troubleshooting

Observation	Possible cause	Recommended action
Low protein binding/yield	Insufficient protein concentration.	Increase the input protein concentration to $\geq \mu\text{g}/\mu\text{L}$ for optimal protein binding before clean-up.
	Insufficient amount of magnetic beads.	Use a minimum of bead slurry based on a 1:25 ratio of bead slurry (μL):protein (μg).
	Insufficient binding time/agitation.	Increase binding time to 45 minutes with sufficient agitation to keep beads in suspension.
Incomplete removal of contaminants	Insufficient washing.	Use Pierce™ Peptide Desalting Spin Columns (Cat. No. 89852) or equivalent to remove excess buffer or TMT™ reagents.
Beads present in output peptide sample	Beads carryover during transfer of supernatant.	Centrifuge collected output peptide samples at $10,000 \times g$ for 2 minutes before LC-MS analysis.
		Use C18 tips or trap columns to remove beads from the output peptide sample before LC-MS analysis.

Related products

Product	Cat. No.
EasyPep™ Magnetic MS Sample Prep Kit	A57867
EasyPep™ Lysis Buffer	A45735
Pierce™ Rapid Gold BCA Protein Assay Kit	A53225
Pierce™ Trypsin/Lys-C Protease Mix, MS Grade	A40007
Pierce™ Peptide Desalting Spin Columns	89852
Pierce™ Quantitative Colorimetric Peptide Assay Kit	23275
Pierce™ Quantitative Fluorometric Peptide Assay	23290

Limited product warranty

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Revision history: Pub. No. MAN0030133 B.0

Revision	Date	Description
B.0	5 January 2024	Correction of typo in Table 1 from 'dilution solution' to 'binding solution'.
A.0	7 November 2023	New document for Pierce™ Proteomics Magnetic Clean-up Beads.

The information in this guide is subject to change without notice.

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